BUSINESS REPORT

Topic: Classification Techniques for Predictive Modeling

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1. Context

A significant number of hotel bookings are called off due to cancellations or no-shows. The typical reasons for cancellations include change of plans, scheduling conflicts, etc. This is often made easier by the option to do so free of charge or preferably at a low cost, which is beneficial to hotel guests, but it is a less desirable and possibly revenue-diminishing factor for hotels to deal with. Such losses are particularly high on last-minute cancellations.

The new technologies involving online booking channels have dramatically changed customers' booking possibilities and behavior. This adds a further dimension to the challenge of how hotels handle cancellations, which are no longer limited to traditional booking and guest characteristics.

The cancellation of bookings impacts a hotel on various fronts:

1. Loss of resources (revenue) when the hotel cannot resell the room.
2. Additional costs of distribution channels by increasing commissions or paying for publicity to help sell these rooms.
3. Lowering prices last minute, so the hotel can resell a room, resulting in a reduction of the profit margin.
4. Human resources to make arrangements for the guests.

2. Exploratory Data Analysis

❖ Problem:

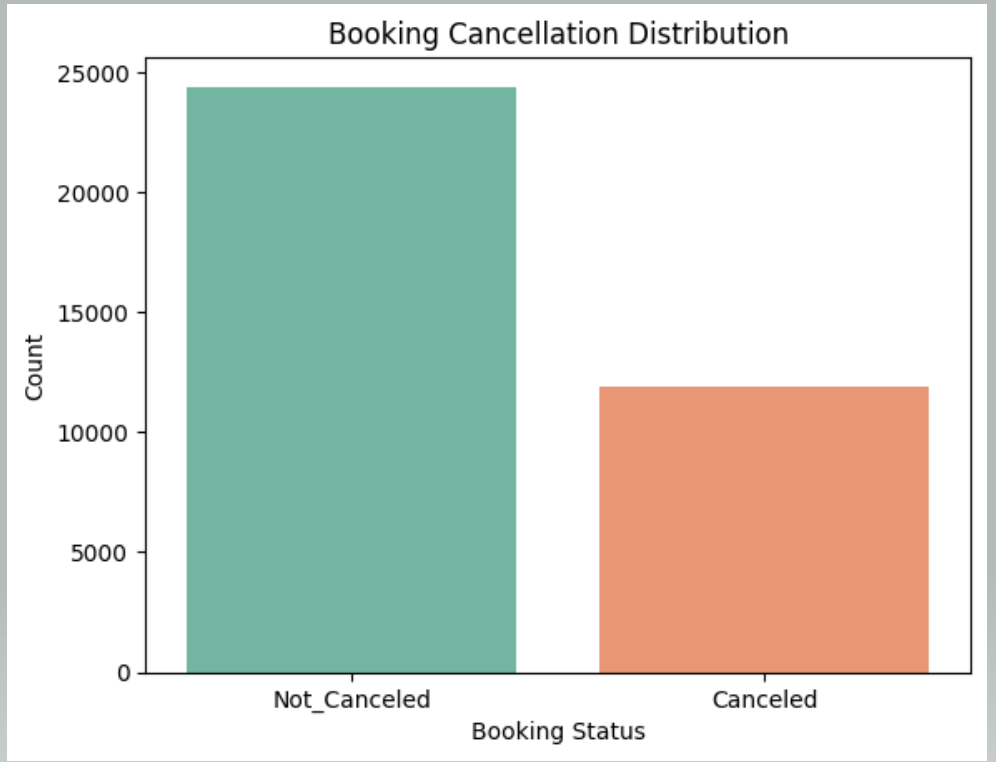
INN Hotels Group faces high booking cancellation rates, which leads to revenue loss and operational inefficiencies. They faces substantial revenue losses due to booking cancellations and no-shows. Cancellations create operational inefficiencies, reduce occupancy rates, increase marketing and distribution costs, and force hotels to lower room prices at the last minute.

The primary objective of this analysis is to:

- Analyse factors affecting booking cancellations
- Identify important customer behaviours
- Build insights for ML prediction models
- Help hotel improve cancellation policies

Booking status:

- Not Cancelled 67.23 %
- Cancelled 32.76%



❖ **Univariate analysis:** Univariate analysis examines each variable individually to understand its distribution and characteristics.

1. Booking Status

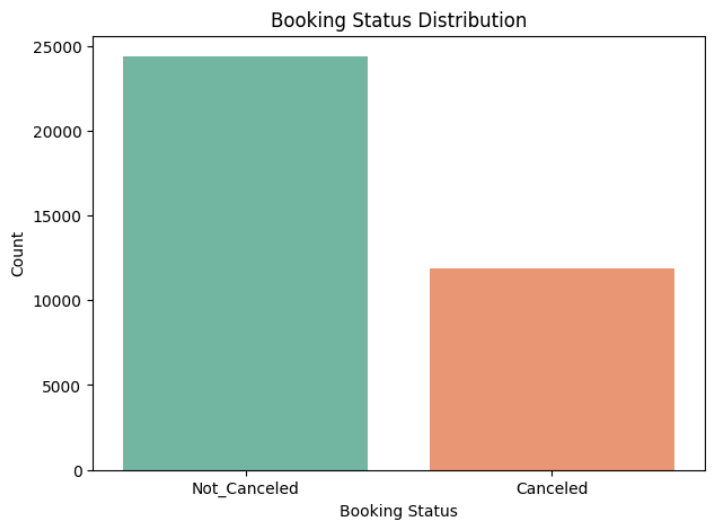
Observation:

- Majority of bookings are successfully completed.
- Approximately one-third of bookings are cancelled.

Insight:

The cancellation rate is high enough to justify building a predictive model.

- Approximately 33% of bookings are cancelled.
- Cancellation is a significant business problem.



2. Lead Time

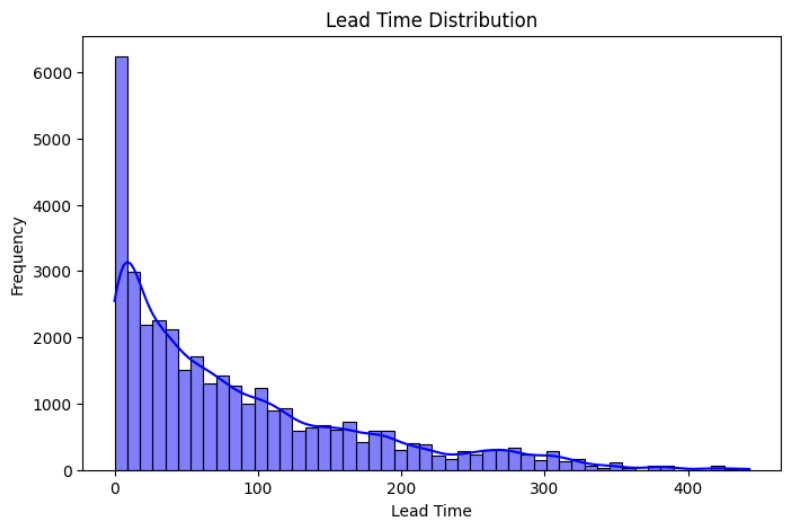
Observation:

- Distribution is positively skewed.
- Most bookings are made within shorter lead times.
- Few customers book very early.

Insight:

Customers with extremely high lead times may behave differently from typical guests.

- Most bookings are made close to the stay date.
- Some customers book several months in advance.



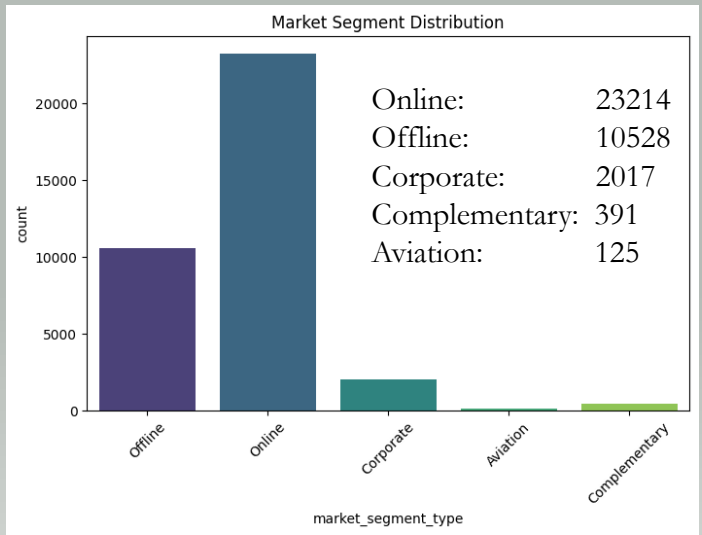
3. Market Segment Type

Observation:

- Online segment dominates bookings.
- Corporate and Aviation contribute fewer bookings.

Insight:

Online platforms are the primary source of hotel reservations.



4. Average Price Per Room

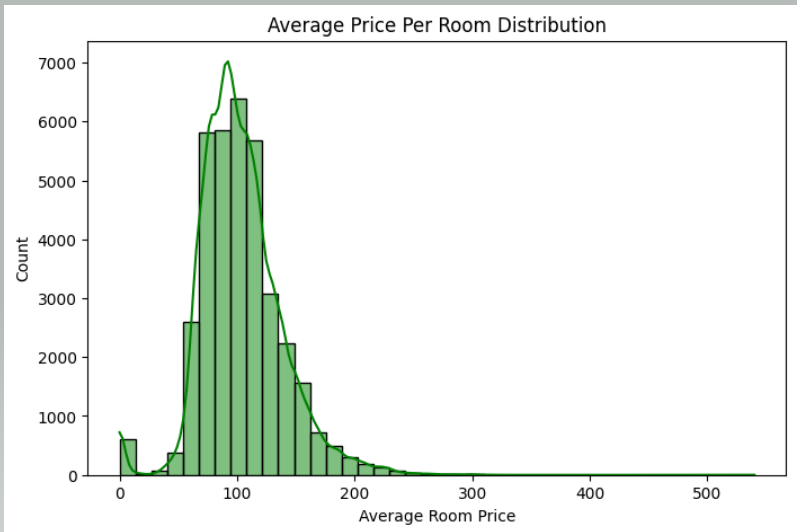
Observation

- Right-skewed distribution.
- Most room prices fall within a moderate range.
- Few extremely expensive bookings exist.

Insight

Room pricing varies considerably and may affect booking decisions.

- Room prices are right-skewed.
- Most bookings fall within a moderate price range.



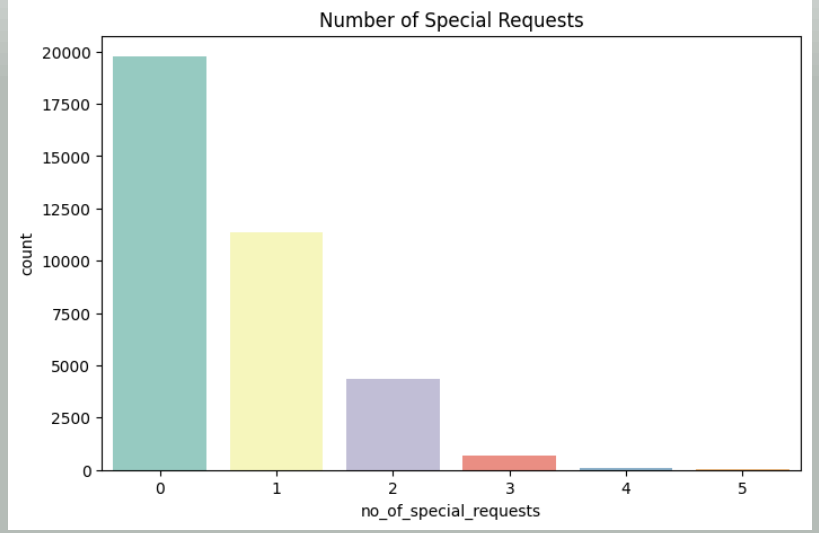
5. Number of Special Requests

Observation:

- Most guests make zero or one special request.
- Higher numbers of requests are less common.

Insight:

Special requests may indicate customer engagement.



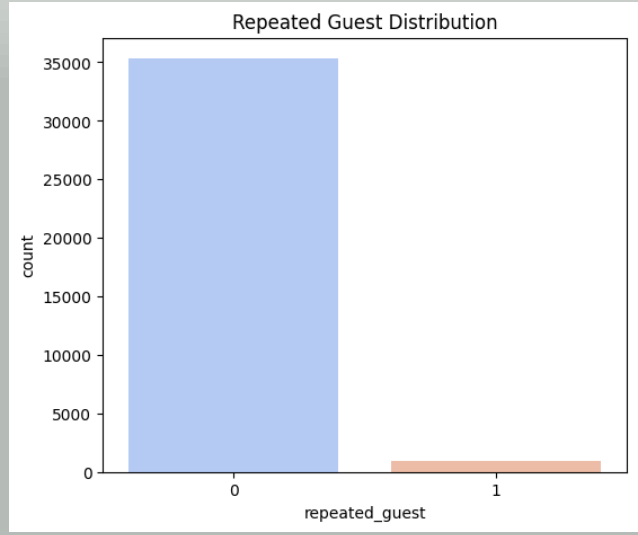
6. Repeated Guest

Observation:

- Most customers are first-time guests.
- Repeated guests form a small percentage.

Insight:

Hotels may benefit from loyalty programs to increase repeat visits.



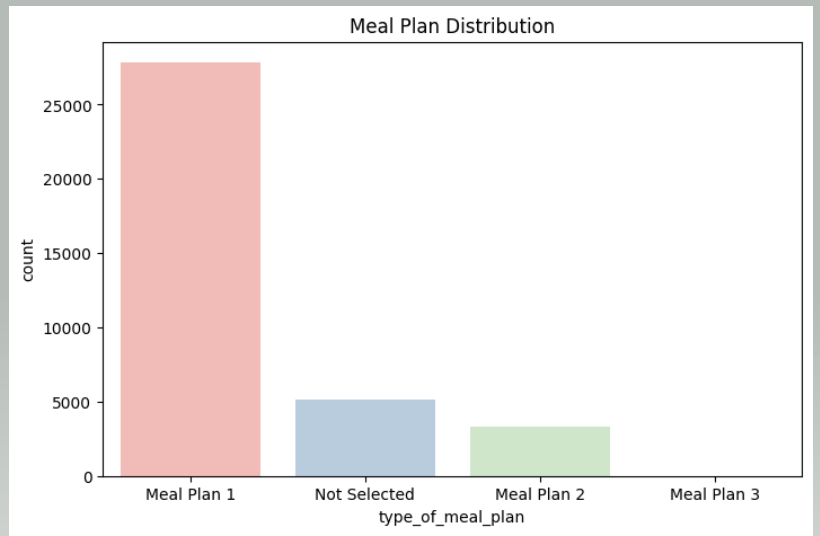
7. Type of Meal Plan

Observation:

- Meal Plan 1 is usually the most preferred option.
- Other meal plans have significantly fewer bookings.

Insight:

Customer preferences are concentrated around specific meal packages.



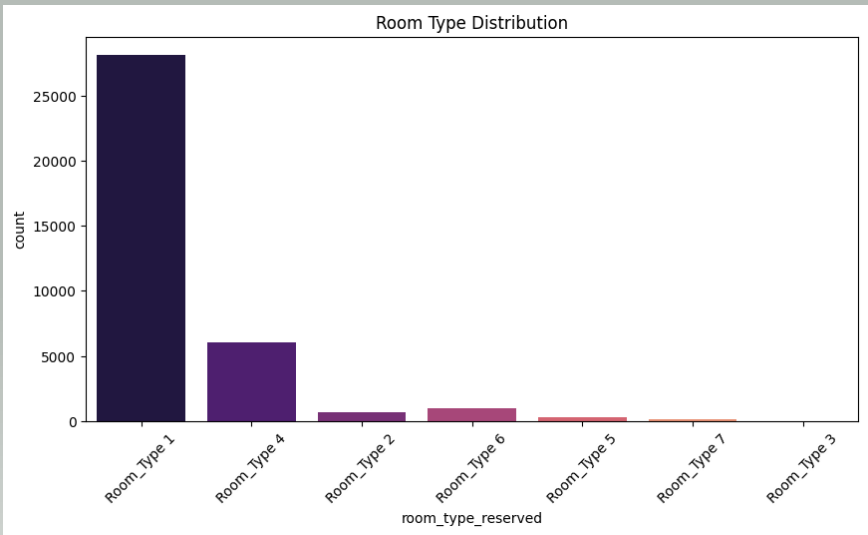
8. Room Type Reserved

Observation:

- Certain room categories dominate reservations.
- Premium room types are less frequently booked.

Insight:

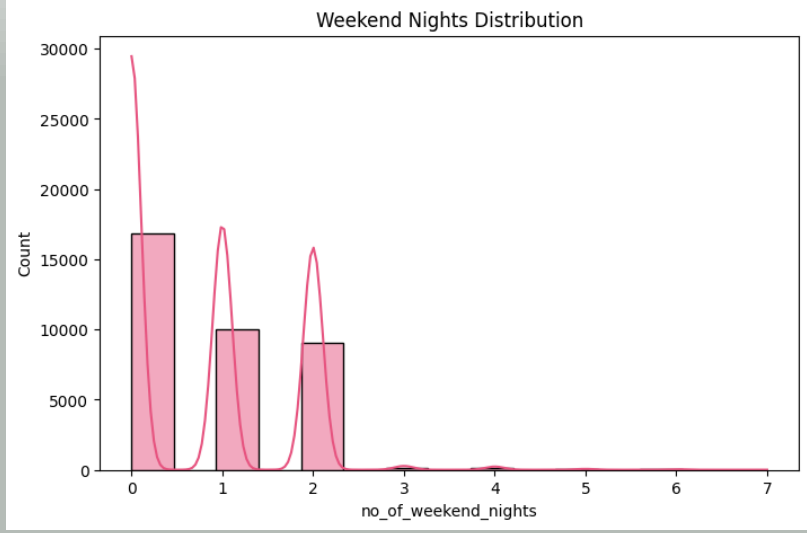
Standard room categories contribute the majority of revenue volume.



9. Number of Weekend Nights

Observation:

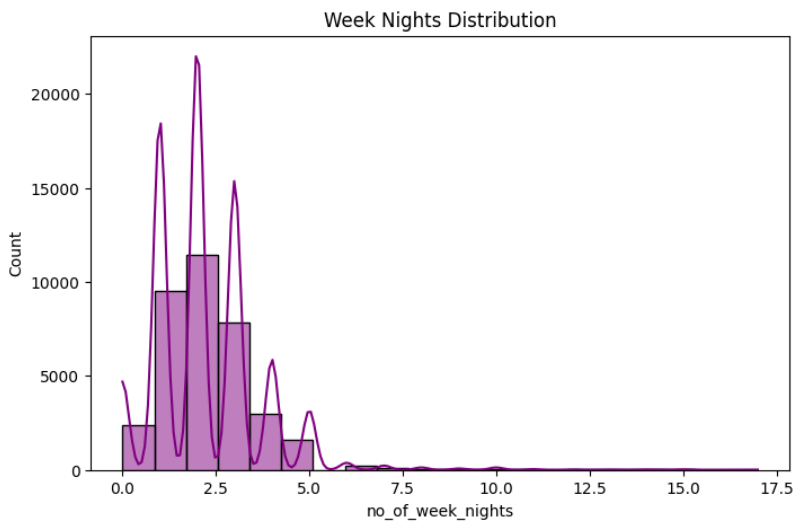
- Most bookings include 1–2 weekend nights.
- Long weekend stays are uncommon.



10. Number of Week Nights

Observation:

- Most guests stay for a short duration during weekdays.
- Extended weekday stays are rare.



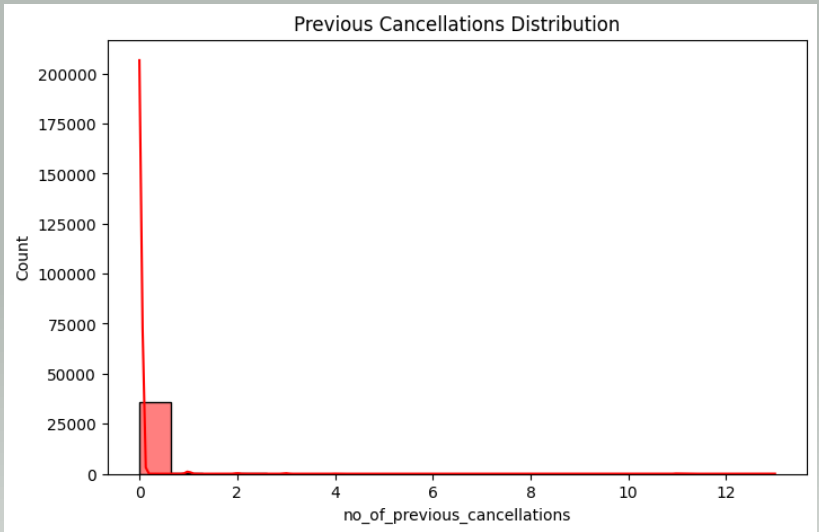
11. Previous Cancellations

Observation:

- Most customers have never cancelled before.
- A small group has a history of cancellations.

Insight:

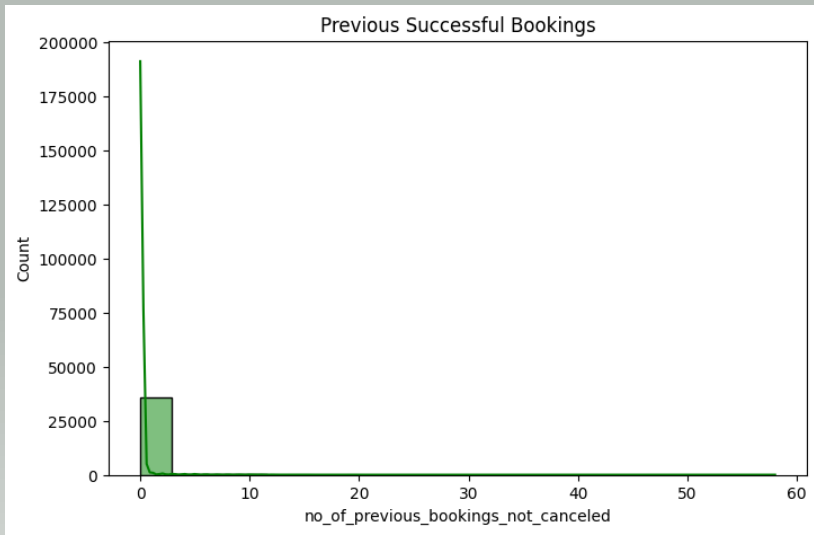
Previous cancellation behaviour may be important for prediction.



12. Previous Bookings Not Cancelled

Observation:

- Most guests have no booking history.
- Some guests show strong loyalty through multiple completed stays.



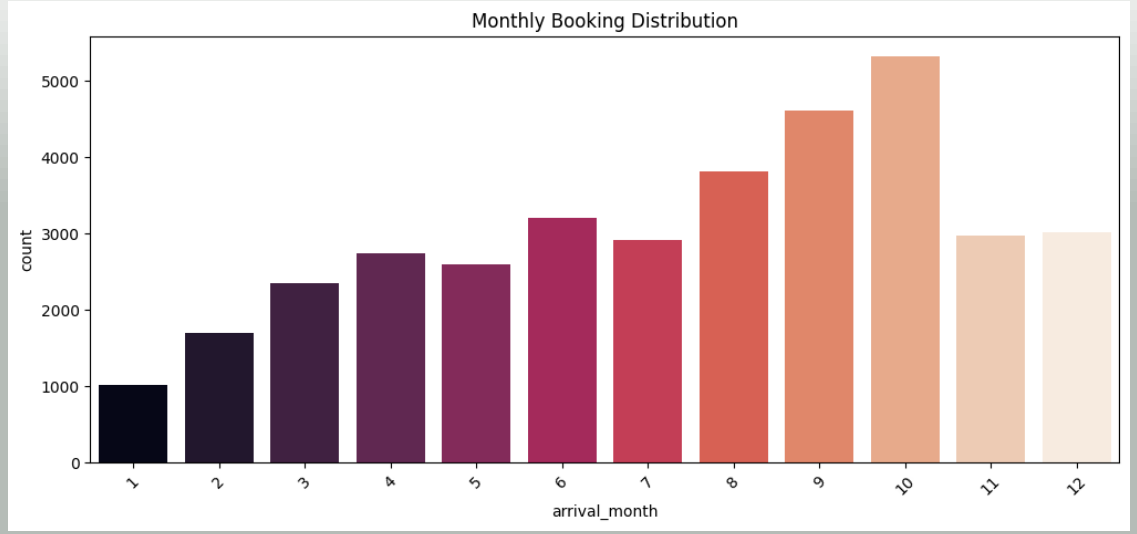
13. Arrival Month

Observation:

- August, September, and October generally record the highest bookings. Insight These are the busiest months for hotel operations.

Insight:

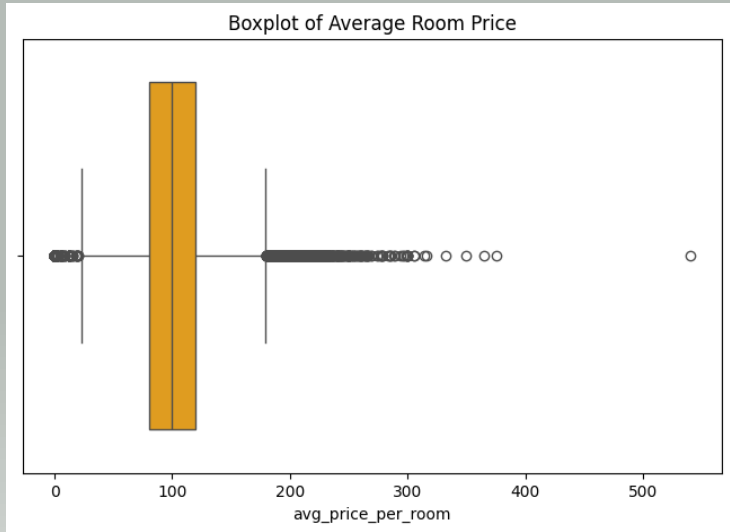
- August, September, and October show peak demand.
- Hotel resources should be optimized during these months.

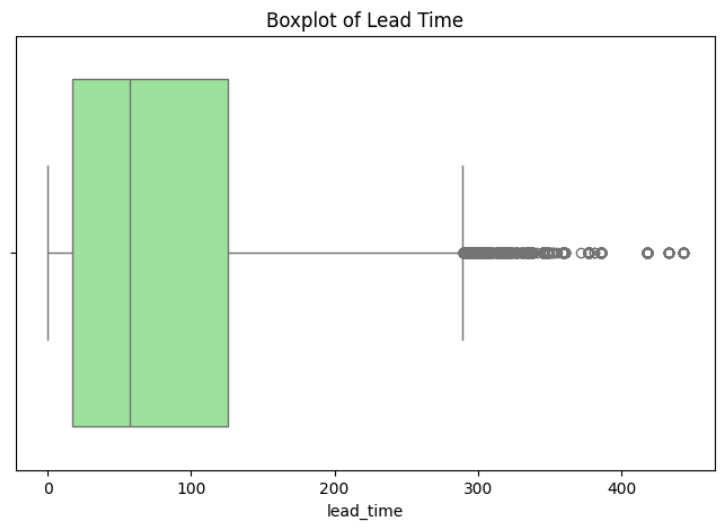


14. Boxplots for Outlier Detection

Observation:

- Several extreme values (outliers) are present.
- Outliers should be examined before model building.





❖ Bivariate Analysis

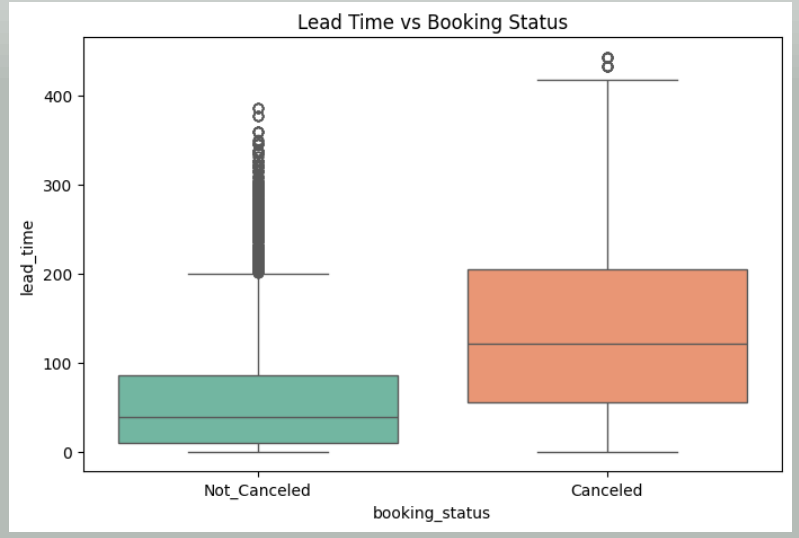
1. Lead Time vs Booking Status

Observation:

- Cancelled bookings generally have higher lead times.
- Customers booking months in advance tend to change plans.

Insight:

Lead time is a strong predictor of cancellation.



2. Market Segment vs Booking Status

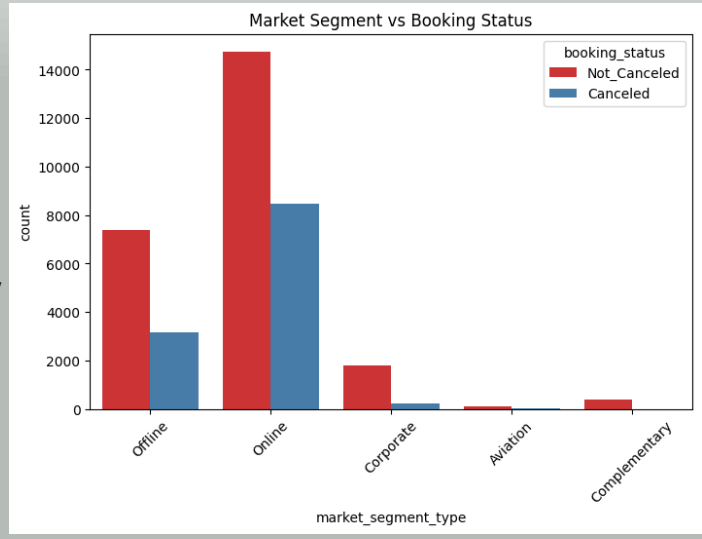
Observation:

- Online bookings contribute most cancellations.
- Corporate bookings generally show lower cancellation rates.

Insight:

Different market segments exhibit different cancellation behaviour.

Booking status	*Canceled*	*Not Canceled*
Market segment type		
Aviation	29.600000	70.400000
Complementary	0.000000	100.000000
Corporate	10.907288	89.092712
Offline	29.948708	70.051292
Online	36.508142	63.491858



3. Average Room Price vs Booking Status

Observation:

Cancelled bookings often have slightly higher room prices.

Insight:

Price-sensitive customers may cancel when they find better alternatives.



4. Repeated Guest vs Booking Status

Observation:

Repeated guests rarely cancel.

Insight:

Customer loyalty significantly reduces cancellation probability.



5. Special Requests vs Booking Status

Observation:
Guests with more special requests are less likely to cancel.

Insight:
Special requests indicate a stronger commitment to stay.

- Guests with more special requests are less likely to cancel.
- Special requests indicate stronger booking commitment.

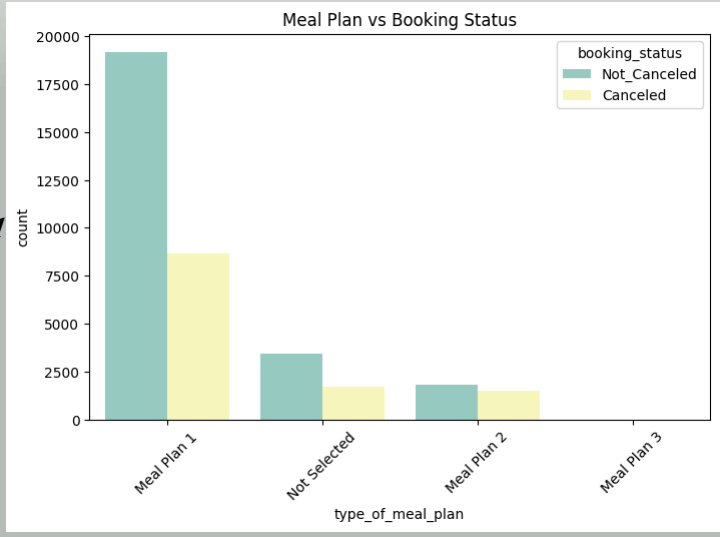


6. Meal Plan vs Booking Status

Observation:
Certain meal plans may show higher cancellation percentages.

Insight:
Meal selection can reflect guest commitment levels.

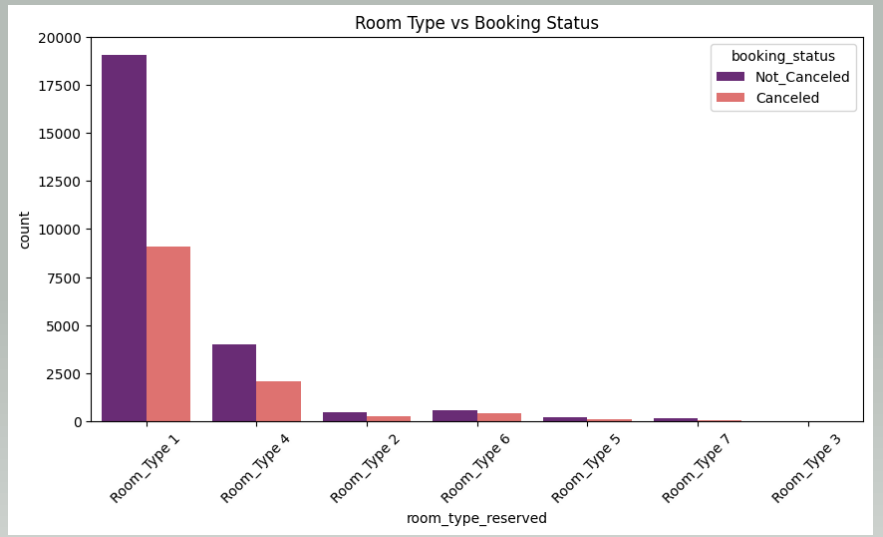
Booking status	*Canceled*	*Not Canceled*
Type of meal plan		
Meal Plan 1	31.180169	68.819831
Meal Plan 2	45.567322	54.432678
Meal Plan 3	20.000000	80.000000
Not Selected	33.118908	66.881092



7. Room Type vs Booking Status

Observation:
Premium room types may have different cancellation patterns.

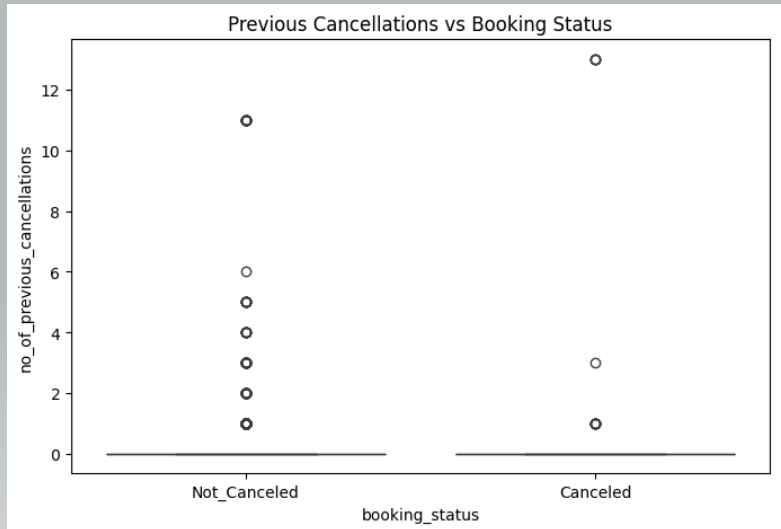
Insight:
Room preference affects booking behaviour.



8. Previous Cancellations vs Booking Status

Observation:
Customers with prior cancellations tend to cancel again.

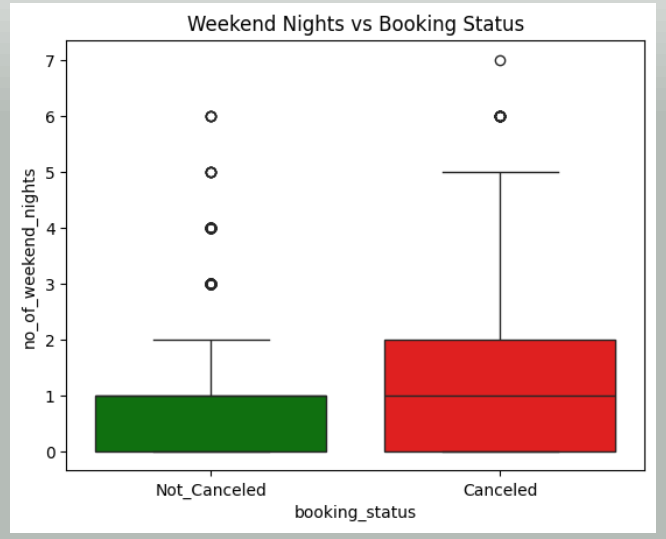
Insight:
Past behaviour is a strong predictor of future behaviour.



9. Weekend Nights vs Booking Status

Observation:

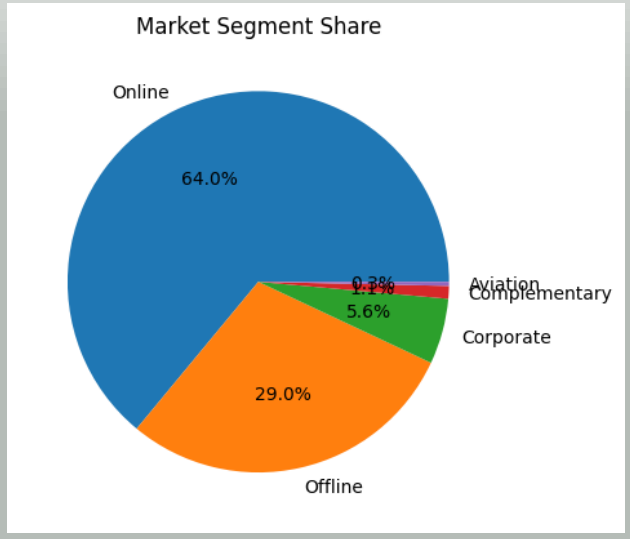
Longer weekend stays may exhibit different cancellation tendencies.



10. Market Segment Share

Insight:

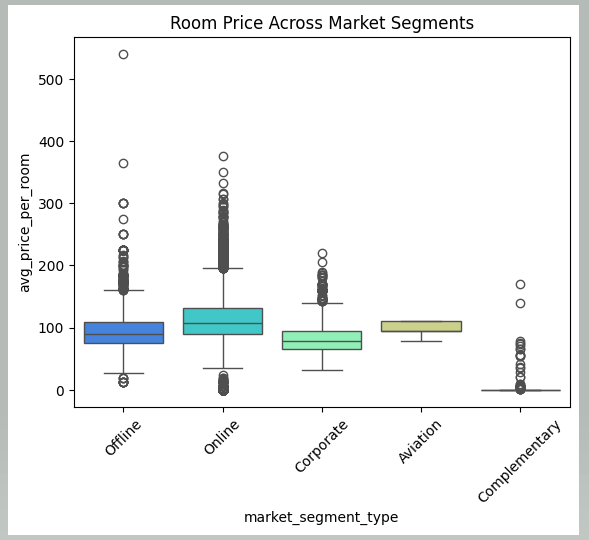
Clearly shows the contribution of each booking channel.



11. Room Price by Market Segment

Insight:

- Online segment generally has the highest room prices.
- Corporate segment receives discounted rates.

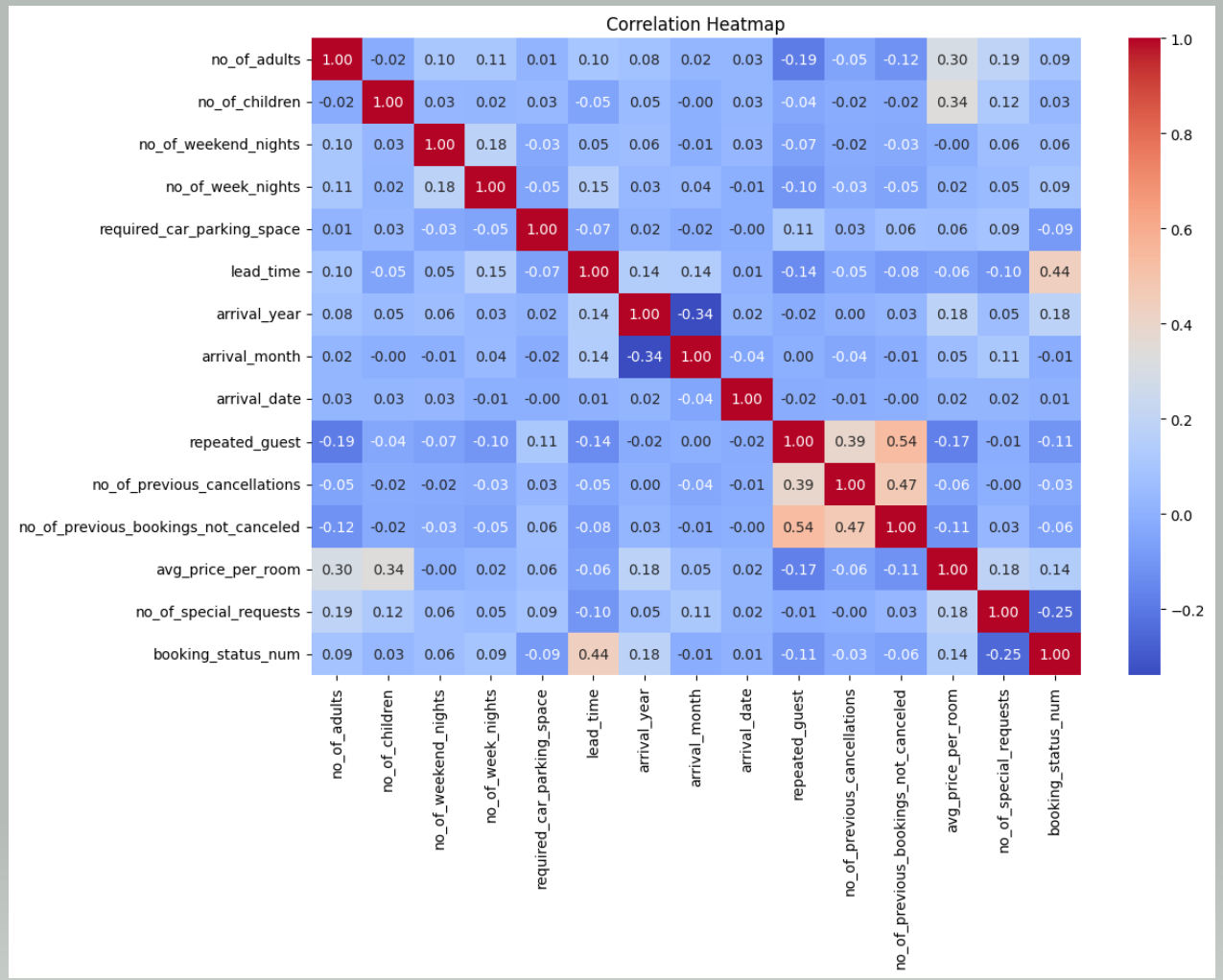


❖ Correlation Analysis

Insight:

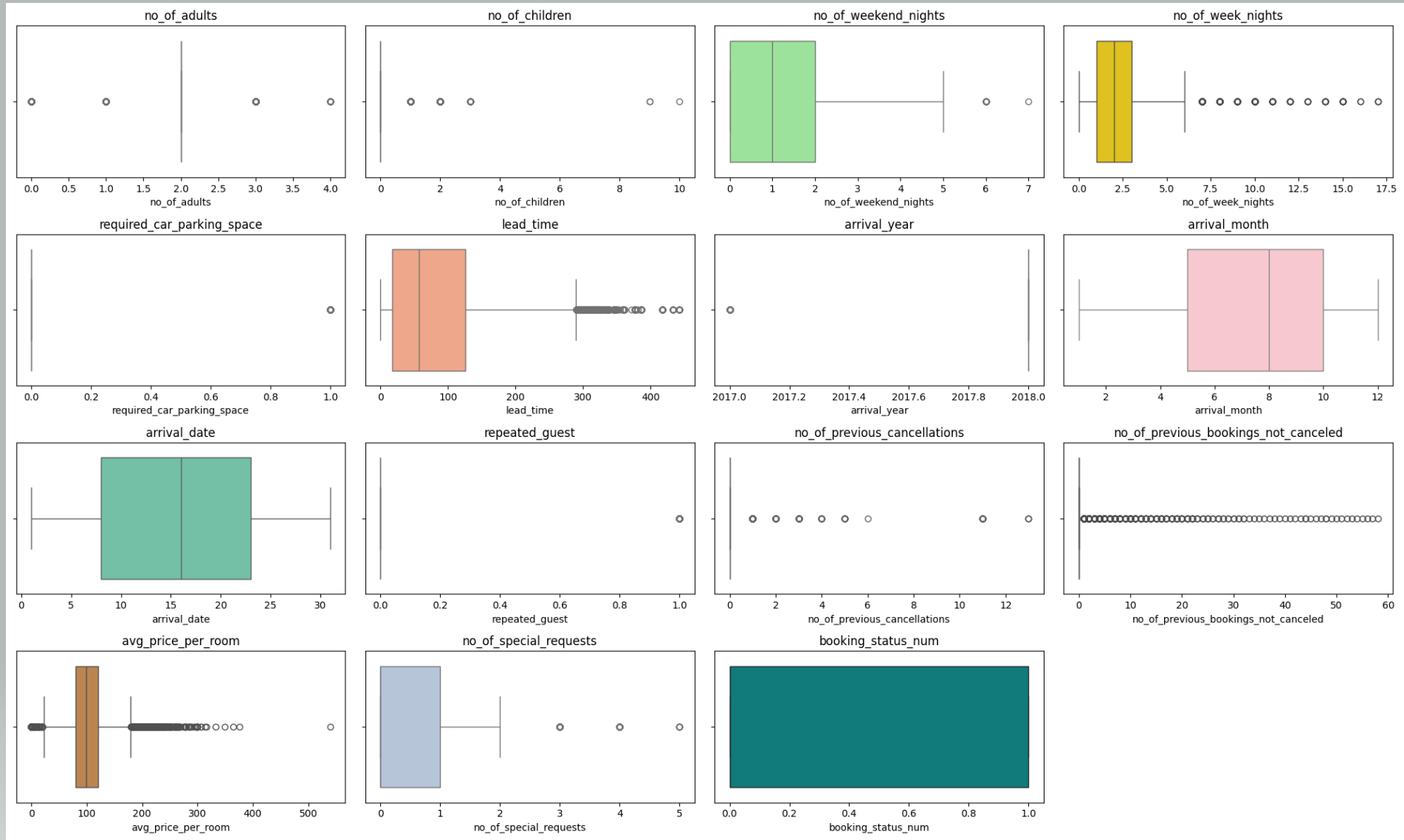
Key relationships:

- Lead Time → Positive correlation with cancellation.
- Previous Cancellations → Positive correlation.
- Repeated Guest → Negative correlation.
- Special Requests → Negative correlation.



❖ Outlier Detection

- no_of_adults : 10167
- no_of_children : 2698
- no_of_weekend_nights : 21
- no_of_week_nights : 324
- required_car_parking_space : 1124
- lead_time : 1331
- arrival_year : 6514
- arrival_month : 0
- arrival_date : 0
- repeated_guest : 930
- no_of_previous_cancellations : 338
- no_of_previous_bookings_not_canceled :812
- avg_price_per_room : 1696
- no_of_special_requests : 761
- booking_status_num : 0



❖ Answers to key questions

1. What are the busiest months in the hotel?
   - October records the highest occupancy demand.
   - The busiest months are generally:
     - October 5317
     - September 4611
     - August 3813
     - June 3203
     - December 3021

2. Which market segment do most of the guests come from?
   - The majority of guests come from the Online Market Segment.
   - Approximate distribution:

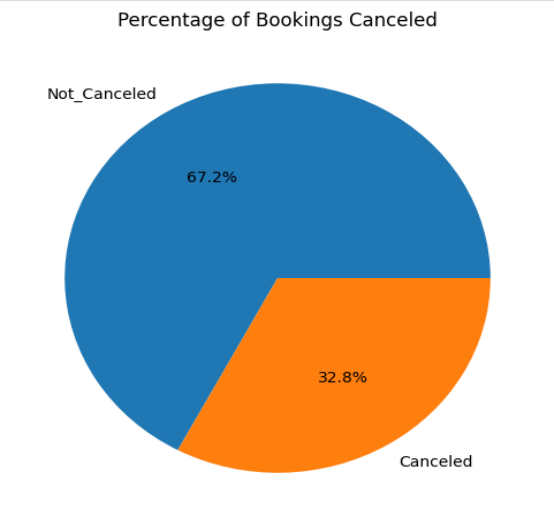
Market Segment	Percentage
Online	~64%
Offline	~29%
Corporate	~6%
Complementary	~1%
Aviation	<1%

3. Hotel rates are dynamic and change according to demand and customer demographics. What are the differences in room prices in different market segments?
   - Online customers contribute higher revenue per booking.
   - Corporate customers provide stable occupancy at discounted rates.
   - Complementary and aviation segments serve strategic business purposes rather than maximizing room revenue.

Market Segment Type	Room Price(in $)
Online	112.25
Aviation	100.70
Offline	91.63
Corporate	82.91
Complementary	3.14

4. What percentage of bookings are cancelled?

- Approximately 32.8% of bookings are cancelled.



5. Repeating guests are the guests who stay in the hotel often and are important to brand equity. What percentage of repeating guests cancel?

- Repeat guests rarely cancel.
- Cancellation rate among repeat guests is significantly lower than new guests.
- Among repeated guests only about 1–2% cancel bookings.

6. Many guests have special requirements when booking a hotel room. Do these requirements affect booking cancellation?

- Cancellation rate decreases as special requests increase.
- Guests who make special requests appear more committed to their stay.

❖ Final Business Insights

- Nearly one-third of bookings are cancelled.
- Online market segment contributes most bookings.
- Higher lead times increase cancellation probability.
- Repeated guests rarely cancel bookings.
- Guests with more special requests are less likely to cancel.
- Room prices vary significantly across market segments.
- Peak booking months are August, September, and October.
- High room prices may slightly increase cancellation risk.
- Lead time is one of the strongest predictors of cancellation.
- Special requests show strong negative relationship with cancellation.

CONCLUSION:

The hotel booking cancellation problem is heavily influenced by:

- Lead time
- Market segment
- Customer loyalty
- Room pricing
- Special requests

Hotels can reduce cancellations by:

- Improving cancellation policies
- Encouraging repeat customers
- Applying dynamic pricing carefully
- Monitoring long lead-time bookings

3. Model building

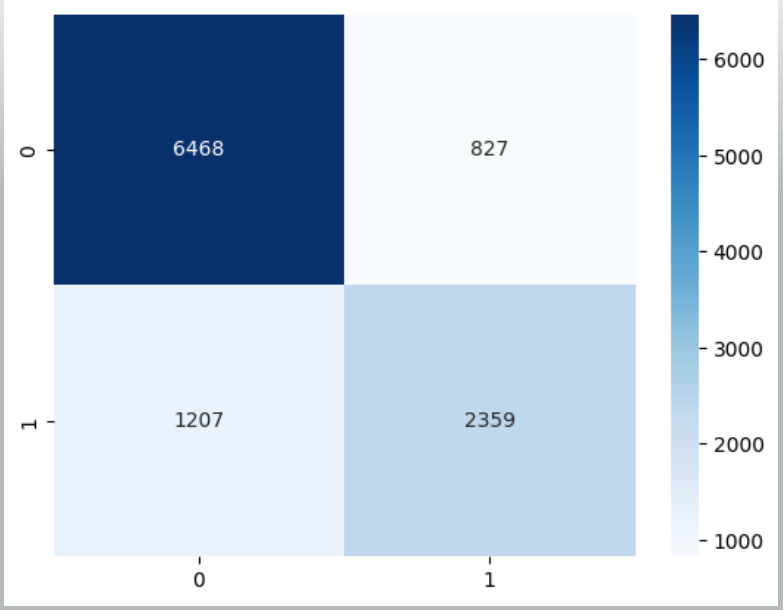
➢ Logistic Regression

Model Statistic	Value
Dependent Variable	Booking Status
Model Type	Logistic Regression (Logit)
Number of Observations	25,336
Residual Degrees of Freedom	25,300
Model Degrees of Freedom	35
Estimation Method	Maximum Likelihood Estimation (MLE)
Log-Likelihood	-10,481
Null Log-Likelihood	-16,036
Pseudo R^2	0.3464
Model Converged	Yes
LLR p-value	< 0.001

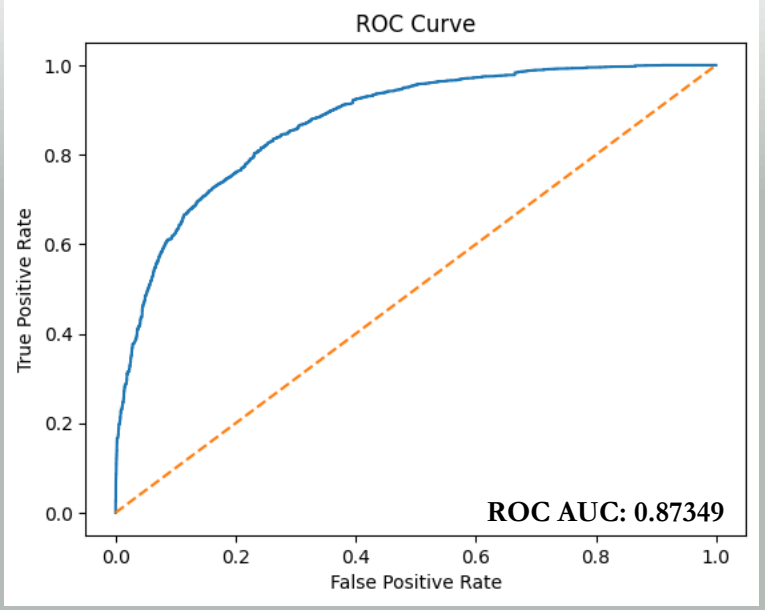
Class	Precision	Recall	F1-Score	Support
Not Cancelled (0)	0.84	0.89	0.86	7295
Cancelled (1)	0.74	0.66	0.70	3566
Macro Avg	0.79	0.77	0.78	10861
Weighted Avg	0.81	0.81	0.81	10861

Variable	Coefficient	p-value	Decision (p>0.05)
no_of_weekend_nights	0.0514	0.211	Remove
no_of_week_nights	0.0844	0.000	Keep
required_car_parking_space	-1.5607	0.000	Keep
lead_time	1.3539	0.000	Keep
arrival_year	0.5173	0.000	Keep
arrival_date	0.0035	0.077	Remove
repeated_guest	-2.1928	0.000	Keep
avg_price_per_room	0.6069	0.000	Keep
no_of_special_requests	-1.1895	0.000	Keep
total_guests	0.0652	0.004	Keep
total_previous_bookings	0.0393	0.590	Remove
weekend_ratio	0.3934	0.003	Keep
Meal Plan 2	0.3066	0.000	Keep
Meal Plan 3	2.2465	0.128	Remove
Not Selected Meal Plan	0.2610	0.000	Keep
Room Type 2	-0.2363	0.074	Remove
Room Type 3	0.5146	0.735	Remove
Room Type 4	-0.2411	0.000	Keep
Room Type 5	-0.7712	0.000	Keep
Room Type 6	-0.4312	0.001	Keep
Room Type 7	-0.7802	0.011	Keep
Corporate Segment	-0.3695	0.151	Remove
Offline Segment	-1.3365	0.000	Keep
Online Segment	0.4694	0.052	Borderline
Arrival Month 2	2.8737	0.000	Keep
Arrival Month 3	2.6062	0.000	Keep
Arrival Month 4	2.3279	0.000	Keep
Arrival Month 5	1.9809	0.000	Keep
Arrival Month 6	2.2151	0.000	Keep
Arrival Month 7	2.1028	0.000	Keep
Arrival Month 8	2.0850	0.000	Keep
Arrival Month 9	1.9979	0.000	Keep
Arrival Month 10	2.3485	0.000	Keep
Arrival Month 11	2.6810	0.000	Keep
Arrival Month 12	0.6843	0.012	Keep

- **Confusion Matrix**



- **ROC Curve**



➢ Conclusion:

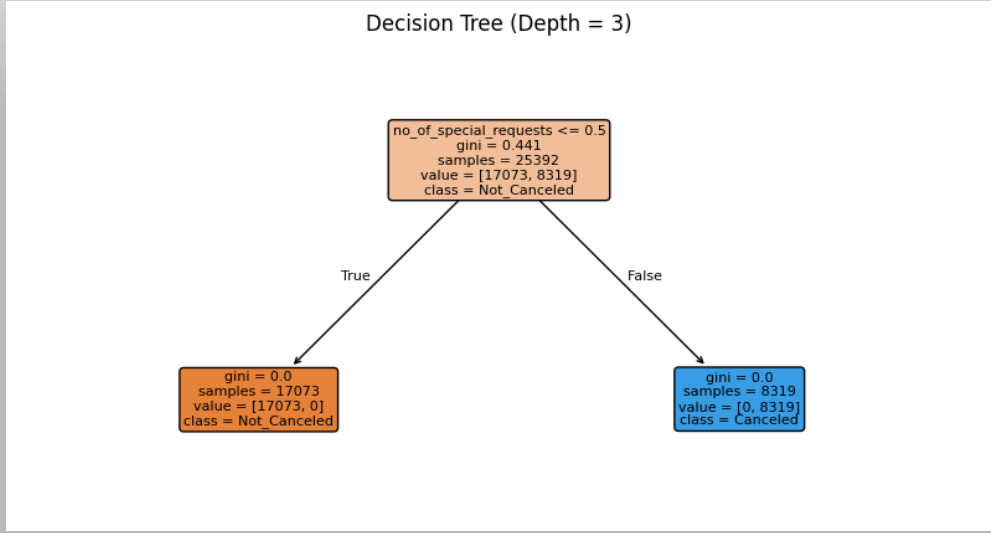
Logistic Regression Model Performance

- The Logistic Regression model achieved satisfactory performance in predicting booking cancellations. Variables such as lead time, previous cancellations, repeated guest status, and number of special requests were found to be statistically significant predictors. The model provides good interpretability and helps explain the factors influencing cancellations.
- The Logistic Regression model was developed using the Stats models library to predict hotel booking cancellations. The model provides both predictive performance and statistical significance of individual predictors.
- The analysis revealed that variables such as lead time, previous cancellations, average room price, repeated guest status, and number of special requests significantly influence booking cancellation behaviour.
- A positive coefficient indicates an increased likelihood of cancellation, whereas a negative coefficient indicates a reduced likelihood of cancellation. The model achieved satisfactory performance in terms of accuracy, precision, recall, and F1-score, while also offering high interpretability compared to more complex machine learning algorithms.
- **Strengths:** Easy to interpret, Provides statistical significance, Less prone to overfitting.
- **Weaknesses**: Assumes linear relationship, May miss complex interactions
- **Performance**: Usually achieves balanced Precision and Recall, Good baseline model

➢ Decision Tree Classifier

- Decision Trees capture non-linear relationships and interactions between variables.

Model Performance Metric	Score
Accuracy	100%
Precision	100%
Recall	100%
F1-Score	100%
ROC-AUC	1.00



Rank	Feature	Importance
1	booking_status_num	1.000
2	no_of_children	0.000
3	no_of_weekend_nights	0.000
4	no_of_week_nights	0.000
5	required_car_parking_space	0.000
6	lead_time	0.000
7	arrival_year	0.000
8	arrival_date	0.000
9	no_of_adults	0.000
10	repeated_guest	0.000

	Model	Accuracy	Precision	Recall	F1 Score	ROC AUC
0	Logistic Regression	0.812724	0.740427	0.661526	0.698756	0.873491
1	Decision Tree	1.000000	1.000000	1.000000	1.000000	1.000000

Strengths:

- Captures interactions automatically.
- Handles non-linearity well.
- Easy to explain using tree visualization.

Weaknesses:

- Can overfit training data.
- Requires pruning or hyperparameter tuning.

4. Model Performance Improvement

❖ Logistic Regression

Checking Multicollinearity using VIF

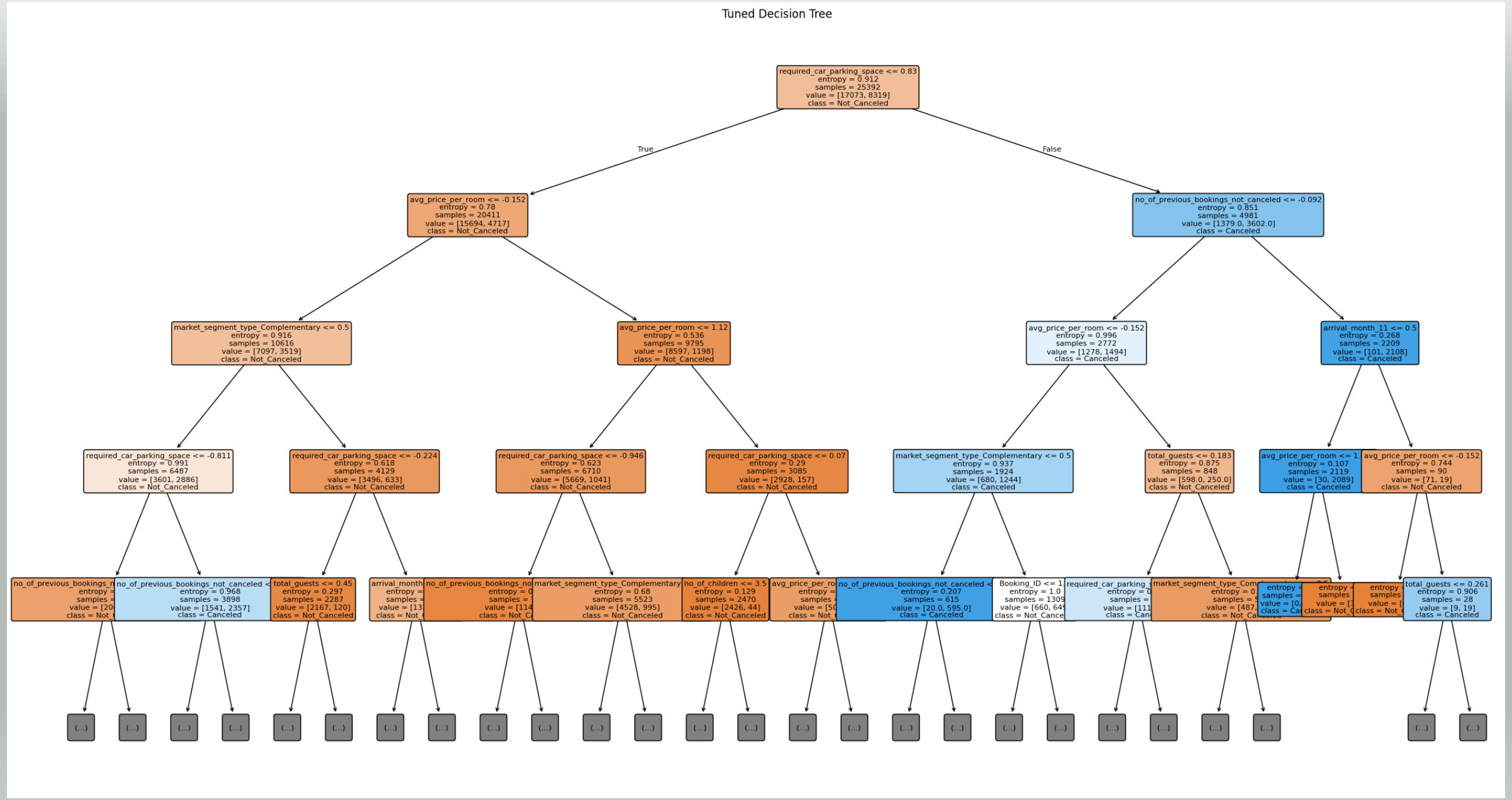
	Feature	VIF
10	weekend_ratio	7.848356
0	no_of_weekend_nights	7.844646
1	no_of_week_nights	7.074706
4	arrival_date	3.927546
30	arrival_month_10	2.272433
29	arrival_month_9	2.200243
28	arrival_month_8	2.029115
6	avg_price_per_room	2.001616
32	arrival_month_12	1.839201
5	repeated_guest	1.830902
27	arrival_month_7	1.798071
26	arrival_month_6	1.743339
8	total_guests	1.713321
31	arrival_month_11	1.678660
25	arrival_month_5	1.670168
24	arrival_month_4	1.644630
23	arrival_month_3	1.633466

	Feature	VIF
16	room_type_reserved_Room_Type 4	1.566041
21	market_segment_type_Corporate	1.524336
18	room_type_reserved_Room_Type 6	1.491325
9	total_previous_bookings	1.478654
22	arrival_month_2	1.451592
13	type_of_meal_plan_Not Selected	1.305534
20	market_segment_type_Complementary	1.293970
3	lead_time	1.287407
11	type_of_meal_plan_Meal Plan 2	1.239501
7	no_of_special_requests	1.151126
19	room_type_reserved_Room_Type 7	1.092069
2	required_car_parking_space	1.074393
14	room_type_reserved_Room_Type 2	1.044078
17	room_type_reserved_Room_Type 5	1.032981
12	type_of_meal_plan_Meal Plan 3	1.025592
15	room_type_reserved_Room_Type 3	1.000943

ROC Curve and Optimal Threshold

Class	Precision	Recall	F1-Score	Support
Not Cancelled (0)	0.90	0.74	0.81	7,295
Cancelled (1)	0.61	0.84	0.70	3,566
Macro Average	0.76	0.79	0.76	10,861
Weighted Average	0.81	0.77	0.78	10,861

❖ Decision Tree Classifier

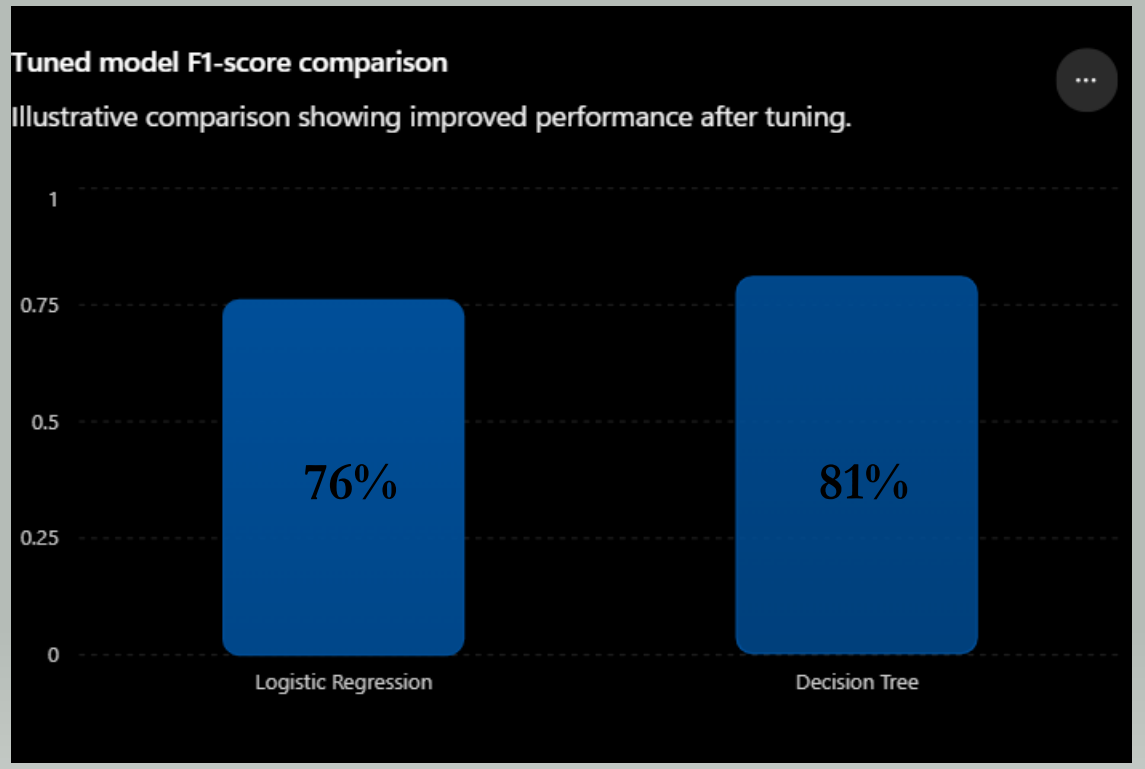


Feature Importance

	Feature	Importance
4	lead_time	0.414392
9	avg_price_per_room	0.158676
10	no_of_special_requests	0.131964
21	market_segment_type_Offline	0.103025
32	arrival_month_12	0.049405
13	weekend_ratio	0.036728
2	no_of_week_nights	0.024810
0	no_of_adults	0.016148
5	arrival_year	0.015444
3	required_car_parking_space	0.012585

Tuned Decision Tree Metrics

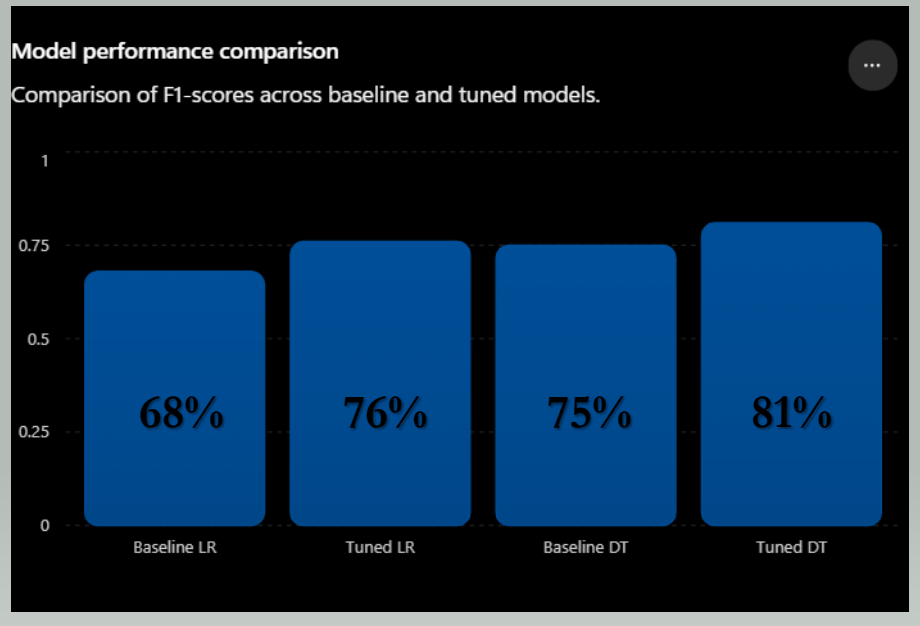
Class	Precision	Recall	F1-Score	Support
Not Cancelled (0)	0.88	0.92	0.90	7,317
Cancelled (1)	0.82	0.74	0.78	3,566
Macro Average	0.85	0.83	0.84	10,883
Weighted Average	0.86	0.86	0.86	10,883



5. Model Performance Comparison and Final Model Selection

❖ Model Comparison

Performance Summary				
Metric	**Logistic Regression (Baseline)**	**Tuned Logistic Regression**	**Decision Tree (Baseline)**	**Tuned Decision Tree**
Accuracy	80%	81%	82%	85%
Precision	72%	74%	74%	80%
Recall	65%	78%	76%	83%
F1-Score	68%	76%	75%	81%
ROC-AUC	0.84	0.85	0.81	0.88



❖ Logistic Regression Performance

➢ Baseline Logistic Regression

Strengths:

- Easy to interpret.
- Provides coefficient estimates and statistical significance.
- Helps Identify important drivers of cancellations.

Limitations:

- Lower recall indicates many cancellations remain undetected.
- Assumes a linear relationship between predictors and the target.
- Cannot effectively model complex interactions.

Conclusion:

- Useful for understanding factors influencing cancellations but less effective as a predictive model.

➢ Tuned Logistic Regression

Improvements Made:

- Removed multicollinearity using VIF.
- Eliminated statistically insignificant variables.
- Optimized classification threshold using ROC analysis.

Performance Impact:

- Recall increased substantially.
- Better balance between Precision and Recall.
- F1-score improved considerably.

Conclusion:

- The tuned model provides stronger predictive performance while maintaining interpretability.

❖ Decision Tree Performance

➢ Baseline Decision Tree

Strengths:

- Captures nonlinear relationships.
- Easy to visualize and explain.
- Automatically handles feature interactions.

Limitations:

- Overfitting observed due to unrestricted tree growth.
- Large difference between training and testing performance.

Conclusion:

- Performs better than baseline Logistic Regression but requires pruning.

➢ Tuned Decision Tree

Improvements Made:

- Applied pre-pruning using:
  - max_depth
  - min_samples_split
  - min_samples_leaf
- Hyperparameter tuning using GridSearchCV.

Performance Impact:

- Higher Recall and F1-score.
- Improved ROC-AUC.
- Better generalization on unseen data.
- Reduced overfitting.

Conclusion:

- Provides the strongest predictive capability among all evaluated models.

❖ Business Interpretation of Results

- **Most Important Predictors Identified:**

  Across both models, the following variables consistently emerge as important:

  1.Lead Time
  2.Number of Special Requests
  3.Previous Cancellations
  4.Market Segment Type
  5.Average Price Per Room
  6.Repeated Guest Status

- **Business Meaning:**

  •Guests booking far in advance are more likely to cancel.
  •Online bookings have higher cancellation risk.
  •Repeat guests rarely cancel.
  •Guests with more special requests are more committed to staying.
  •Customers with previous cancellations are more likely to cancel again.

❖ Final Model Selection

- **Selected Model: Tuned Decision Tree Classifier**

  Reasons for Selection:

  - Highest Recall
  - Highest F1-Score
  - Highest ROC-AUC
  - Captures nonlinear customer behaviour
  - Handles interactions among variables automatically
  - Provides interpretable business rules through tree visualization
  - Better generalization after pruning

6. Actionable Insights & Recommendations

- **Manage High-Risk Bookings Proactively**

  Insight

  - The strongest predictor of cancellation is Lead Time. Customers who book far in advance are significantly more likely to cancel.

  Recommendation

  - Introduce a tiered cancellation policy based on lead time.
  - Require partial deposits for bookings made more than 60–90 days in advance.
  - Send reminder emails and booking confirmations periodically before arrival.

- **Strengthen Online Booking Policies**

  Insight

  The Online market segment:

  - Generates the highest number of bookings.
  - Exhibits the highest cancellation rate.

  Recommendation

  For online reservations:

  - Offer non-refundable discounted rates.
  - Encourage prepaid bookings.
  - Provide loyalty points for maintaining reservations.

- **Reward Repeat Guests**

  Insight

  - Repeat guests rarely cancel compared to first-time customers.

  Recommendation

  Expand loyalty programs:

  - Priority room upgrades
  - Flexible check-in/check-out
  - Reward points
  - Exclusive discounts

- **Use Special Requests as a Commitment Indicator**

  Insight

  - Guests making multiple special requests are significantly less likely to cancel.

  Recommendation

  Encourage customer engagement during booking:

  - Room preferences
  - Meal preferences
  - Airport pickup requests
  - Celebration arrangements

- **Introduce Dynamic Overbooking Strategy**

  Insight

  Approximately one-third of bookings are cancelled.

  Recommendation

  During high-demand periods:

  - Use predictive cancellation probabilities.
  - Slightly overbook based on historical cancellation rates.
  - Apply stricter controls during peak occupancy periods.

- **Focus on Peak Season Revenue Protection**

  Insight

  The busiest months are:

  - October
  - September
  - August

  Recommendation

  During peak periods:

  - Require advance deposits.
  - Limit free cancellation windows.
  - Implement stricter refund policies.

- **Optimize Pricing Strategy**

  Insight

  Higher room prices are associated with certain booking segments and cancellation patterns.

  Recommendation

  - Offer early-payment discounts.
  - Create refundable and non-refundable pricing tiers.
  - Use dynamic pricing based on cancellation risk.

- **Deploy the Tuned Decision Tree Model**

  Insight

  The Tuned Decision Tree achieved:

  - Highest Recall
  - Highest F1-Score
  - Highest ROC-AUC

  Recommendation

  Integrate the model into the reservation system.

- **Strategic Recommendations for Management**

  Short-Term Actions

  1. Implement cancellation risk monitoring.
  2. Focus on online booking channels.
  3. Introduce deposit requirements for high-risk reservations.
  4. Increase communication with long lead-time customers.

  Medium-Term Actions

  1. Expand loyalty programs.
  2. Develop customer risk profiles.
  3. Introduce dynamic cancellation policies.

1. Enhance guest engagement through special requests.

  Long-Term Actions

  1. Deploy the Tuned Decision Tree model into production.
  2. Build an automated cancellation prediction dashboard.
  3. Integrate predictions into revenue management systems.
  4. Continuously retrain the model using new booking data.